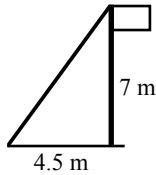




Pythagoras' Theorem. Worded Questions



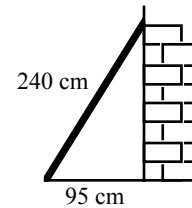
1).



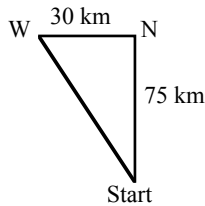
A girl guide ties a wire to a flag pole to secure it. The flag pole is 7 metres long and she is to anchor it 4.5 metres away from the base of the flag pole. How long must her wire be in order to do this?

2).

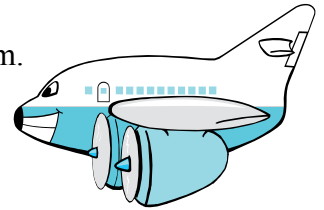
A man has a ladder 240 cm long. He places it against a wall. For safety, the bottom of the ladder must be 95 cm from the base of the wall. How high up the wall will the ladder reach?



3).

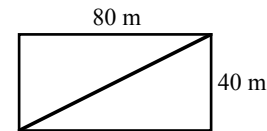


A plane flies due North for 75 km then due West for 30 km. How far is the plane from the original starting position?



4).

A hockey pitch is 80 metres by 40 metres. Mandy knocks the ball from one corner flag to the corner flag diagonally opposite. How far has the ball travelled?



5).

A square has sides of length 32 cm. Find the length of one of its diagonals.

6).

A sports-fan finds out the dimensions of various sports-pitches. Find the **diagonal** distance of each of the following.

- A basketball court is 50 feet wide by 94 feet long.
- An American football pitch is 160 feet wide by 300 feet long.
- A snooker table is 4.5 feet wide by 9 feet long.
- A squash court is 18.5 feet wide by 32 feet long.
- A volleyball court is 29.5 feet wide by 59 feet long.
- A tennis court is 36 feet wide by 78 feet long.
- A lacrosse pitch is 60 yards wide by 110 yards long.
- A table-tennis table is 5 feet wide by 9 feet long.
- An ice hockey pitch is 85 feet wide by 200 feet long.



7).

A ladder reaches 6.8 metres up a vertical wall. The base of the ladder is on horizontal ground, 1.8 metres away from the bottom of the wall. How long is the ladder?

8).

An orienteering course is set out as follows. From the start travel 6 km due South, then 4 km due East and then return to the start. Find

- the distance of the return (last) leg,
- the total distance you would run if you completed the whole course from start to finish.

9).

Calculate the distance between the co-ordinates.

- a. (3,5) and (8,17) b. (0,0) and (3,4) c. (3,5) and (4,9)

10).

The size of Premiership football pitches are determined by this F.A. rule. 'The width of the pitch must lie between 50 yards and 100 yards, the length of the pitch must lie between 100 yards and 130 yards'. Find the **biggest** and **smallest** diagonal you could make on a Premiership football pitch.

